

Tran Manh Duy

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Summary

Systems-oriented Rust developer with a strong foundation in Computer Science and low-level programming. Experienced in building high-performance, low-latency services with a deep understanding of systems programming concepts including memory management, concurrency, and inter-process communication (gRPC, message queues). Proficient in designing modular, trait-based architectures and real-time event processing pipelines using Kafka and async Rust (Tokio). Eager to apply systems-level expertise to security infrastructure and real-time telemetry platforms.

Education

University of Engineering and Technology, VNU

BS in Information Technology, CPA: 3.0/4.0

Available for Full-time employment

Hanoi, Vietnam

Sept 2022 – Sept 2026

Skills

Programming Languages: Rust (Advanced), Golang, Typescript, Java

Frameworks: Axum, Actix, Tokio

Database & Caching: PostgreSQL, MySQL, MongoDB, Redis, SeaOrm, DieselOrm

System & Infrastructure: Docker, Kubernetes, Kafka, RabbitMQ, Git/GitHub, Linux

Cryptography & Blockchain: ZKP (zk-SNARKs, zk-STARKs), Smart Contracts (Solidity/Rust), Layer 2 Protocols

Language: English, Japanese

Other: Document writing, Researching, Self-learning, Communication, Teamwork

Experience

Sotatek ., JSC

Hanoi, Vietnam | Jan 2024 – Present

Rust Developer

- Designed and implemented high-performance **RESTful APIs** (using Axum/Actix) and **gRPC services** (using Tonic, Prost) to power CEX/DEX platforms, serving both web and mobile interfaces with low latency.
- Engineered scalable asynchronous messaging systems using **Kafka** and **RabbitMQ** to handle high-frequency trading data and real-time blockchain event processing.
- Optimized database performance and data reliability by integrating **Redis** for caching and **SeaOrm** for type-safe interactions with PostgreSQL.

Blockchain Developer

- Developed and deployed a Starknet Layer 2 solution integrated with Aptos Layer 1, including data availability on Move, Madara chain configuration, and a Rust-based prover for on-chain verification.

Research Assistant

- Researched, documented, and implemented algorithms related to Zero-Knowledge Proofs (ZKPs), including zk-SNARKs and zk-STARKs.

High-Performance E-commerce Microservice

- **Stack:** Rust (Axum, Tokio), PostgreSQL (SeaOrm), Redis, Docker, GitHub Actions, k6.
- Architected a scalable backend system using **Clean Architecture** and **Workspace** pattern, decoupling business logic (**core**) from infrastructure (**infra**) for maintainability.
- Solved critical **race conditions** in inventory management ensuring data integrity under high concurrency (verified via rigorous integration tests).
- Optimized API latency by implementing **Redis caching** strategies and conducting **load testing (k6)** to ensure system stability under heavy traffic.
- Established a complete **CI/CD pipeline** using GitHub Actions for automated testing and Docker containerization.

Advanced Cryptography & Zero-Knowledge Suite

- **Stack:** Rust, Monorepo, Finite Fields (NTT), Elliptic Curves (BN254), STARKs.
- Architected a **modular monorepo** to implement cryptographic primitives from the ground up, separating core logic (mathlib, curvelib) from protocol implementations.
- Developed **mathlib**: A high-performance library featuring BigInt arithmetic, Finite Fields (Montgomery reduction), and **Number Theoretic Transforms (NTT)** for fast polynomial operations.
- Built **curvelib**: Implemented Elliptic Curve operations (focusing on **BN254**) and Bilinear Pairings required for modern ZK-SNARK protocols.
- Implementing **ZK-STARK** systems, specifically constructing **FRI commitment schemes** and Algebraic Intermediate Representation (AIR).